

Amazon Aurora – Key Concepts (My Notes)

- **Aurora Storage Auto-Scaling**

Aurora automatically increases storage as the database grows. Storage expands in **10 GB increments** and can scale up to **128 TB**, so there is no need to manually provision extra storage in advance.

- **Built-in High Availability**

Aurora automatically keeps **6 copies of data across 3 Availability Zones (AZs)**. This provides durability and fault tolerance without requiring additional configuration from me.

- **Writer Endpoint vs Reader Endpoint**

The **Writer Endpoint** is used for read and write operations such as INSERT, UPDATE, and DELETE. The **Reader Endpoint** distributes read-only queries across Aurora replicas, helping to improve application performance and reduce load on the primary database.

- **Aurora Serverless v2**

Aurora Serverless v2 automatically scales database compute capacity up or down based on workload demand. Unlike traditional scaling, it adjusts in fine-grained increments, allowing the database to respond quickly to changing traffic while optimizing cost.

- **Aurora Global Database**

Aurora Global Database enables a single database to span multiple AWS Regions. Data written in the primary Region is replicated to secondary Regions with **typically less than one second of replication lag**, making it ideal for global applications and disaster recovery scenarios.

My Takeaway

Aurora separates compute from storage, giving it strong scalability, high availability, and global replication capabilities. It is designed to reduce operational overhead while maintaining enterprise-grade performance and resilience.